HR Employee Analytics

Southwest Technology Solutions

100 pts

Unit 3 — Excel Spreadsheets · Core Project Option B

/ 100 points · 40 tasks

How to use this instruction sheet: Download the starter file from the Project Grader, open it in Microsoft Excel 365, and keep this sheet beside you. Work through each task in order. Each task tells you what to do, why it matters, where to find it, and the correct formula syntax — but you must enter the formula yourself. The starter file contains clean data only — no instructions are embedded in the file itself.

E1 — Manage Worksheets & Workbooks

18 points (8 tasks)

2

Task 1 — Add a new worksheet to the workbook and rename it "Org Chart"

WHY THIS MATTERS

Organizing a workbook with multiple named sheets is a professional standard. Each sheet serves a distinct purpose and is clearly labeled.

WHERE TO FIND IT

Right-click any sheet tab at the bottom of Excel → Insert → Worksheet. Then double-click the new tab and type "Org Chart". Press Enter.

HOW YOU'LL KNOW IT WORKED

A new sheet tab labeled "Org Chart" appears at the bottom of the workbook.

2

Task 2 — Freeze row 2 and column A so employee names and headers stay visible

WHY THIS MATTERS

With 30 employees and many columns, freezing keeps the header row and name column visible as you scroll.

WHERE TO FIND IT

Click cell B3. View tab → Freeze Panes → Freeze Panes.

HOW YOU'LL KNOW IT WORKED

Dark freeze lines appear below row 2 and to the right of column A.

2

Task 3 — Navigate to the named range "SalaryData" using the Name Box

WHY THIS MATTERS

Named ranges let you jump to important data instantly. This confirms the range was defined correctly.

WHERE TO FIND IT

Click the Name Box → type SalaryData → press Enter.

HOW YOU'LL KNOW IT WORKED

The salary column cells are selected after typing SalaryData.

2

Task 4 — Configure print settings: set print area and add row/column headers to print

WHY THIS MATTERS

Printing row and column headers helps readers cross-reference printed data back to the spreadsheet.

WHERE TO FIND IT

Select the data range. Page Layout → Print Area → Set Print Area. Then Page Layout → Page Setup dialog → Sheet tab → check Row and column headings.

HOW YOU'LL KNOW IT WORKED

Print Preview shows column letters and row numbers along the edges.

2

Task 5 — Insert a new row at top for the company title; hide the EmpID column from view

WHY THIS MATTERS

A title row makes the workbook self-explanatory. Hiding internal ID columns reduces visual clutter for reports shared with managers.

WHERE TO FIND IT

Right-click row 1 → Insert. Type the company title in A1. Then right-click the EmpID column header → Hide.

HOW YOU'LL KNOW IT WORKED

A title row appears at top. The EmpID column disappears but is not deleted.

2

Task 6 — Apply the "Facet" workbook theme

WHY THIS MATTERS

The Facet theme uses clean modern typography appropriate for professional HR documents.

WHERE TO FIND IT

Page Layout tab → Themes group → Themes → find and click Facet.

HOW YOU'LL KNOW IT WORKED

The workbook fonts and colors update to the Facet theme palette.

2

Task 7 — Set workbook properties: Title = "STS HR Analytics 2026", Author = your name

WHY THIS MATTERS

HR workbooks contain sensitive data. Proper document properties identify the file owner and purpose for compliance.

WHERE TO FIND IT

File → Info → Properties panel (right side) → fill in Title and Author fields.

HOW YOU'LL KNOW IT WORKED

File → Info shows the correct Title and Author.

4

Task 8 — Save the workbook as both .xlsx and .csv formats

WHY THIS MATTERS

.xlsx preserves all formatting and formulas. .csv is a plain data export compatible with HRIS systems and payroll software.

WHERE TO FIND IT

Save as .xlsx normally (Ctrl+S). Then File → Save As → change type to CSV (Comma delimited) → Save.

HOW YOU'LL KNOW IT WORKED

Two files exist in your folder — one .xlsx and one .csv.

E2 — Manage Data Cells & Ranges

18 points (8 tasks)

2

Task 9 — Use AutoFill to extend the EmpID series E001, E002 through E030

WHY THIS MATTERS

AutoFill recognizes the pattern E001, E002 and continues the sequence automatically.

WHERE TO FIND IT

The starter file has E001 in row 3 and E002 in row 4. Select both. Drag the fill handle down through row 32.

HOW YOU'LL KNOW IT WORKED

Cells show E001 through E030 in order down column A.

2

Task 10 — Insert three new columns after Employee Name for Department, Level, and Start Date

WHY THIS MATTERS

Proper data structure separates different information into its own column. Department, Level, and Start Date each need their own column.

WHERE TO FIND IT

Right-click the Salary column header → Insert (repeat 3 times). Label the new columns Department, Level, Start Date. Enter the data from the PDF instruction sheet.

HOW YOU'LL KNOW IT WORKED

Three labeled columns appear between Employee Name and Salary with data for all 30 employees.

2

Task 11 — Format the Salary column as Currency and the Start Date column as Short Date

WHY THIS MATTERS

Currency formatting adds $ and commas. Short Date shows dates in MM/DD/YYYY instead of a raw number.

WHERE TO FIND IT

Select Salary column → Home → $ Currency. Select Start Date column → Home → Number dropdown → Short Date.

HOW YOU'LL KNOW IT WORKED

Salaries show $92,000.00 style. Dates show 3/15/2019 style.

2

Task 12 — Merge and center the company header across A1:G1

WHY THIS MATTERS

A merged title spanning all columns creates a professional header identifying the workbook.

WHERE TO FIND IT

Select A1:G1. Home → Merge & Center. Type the company title.

HOW YOU'LL KNOW IT WORKED

A1:G1 appears as one wide cell with the title centered.

2

Task 13 — Wrap text in the Notes column so long entries are fully visible

WHY THIS MATTERS

Without wrapping, long text is cut off. Wrapping makes all content readable without widening columns excessively.

WHERE TO FIND IT

Select the Notes column. Home tab → Alignment group → Wrap Text button.

HOW YOU'LL KNOW IT WORKED

Notes cells expand to show full text on multiple lines.

3

Task 14 — Apply conditional formatting to highlight salaries above $80,000 in green

WHY THIS MATTERS

Highlighting high earners helps HR quickly identify senior-level compensation.

WHERE TO FIND IT

Select the Salary column. Home → Conditional Formatting → Highlight Cells Rules → Greater Than → 80000 → Green Fill.

HOW YOU'LL KNOW IT WORKED

All salary cells above $80,000 have a green background.

2

Task 15 — Use Format Painter to copy header row formatting to the totals row

WHY THIS MATTERS

Consistent formatting between header and totals rows makes the table look balanced and intentional.

WHERE TO FIND IT

Click any header row cell. Home → Format Painter (double-click to lock). Click and drag across the totals row. Press Escape.

HOW YOU'LL KNOW IT WORKED

The totals row has the same bold text, background color, and borders as the header row.

3

Task 16 — Define named range "SalaryData" for all salary values

WHY THIS MATTERS

Naming the salary range enables readable formulas like =SUM(SalaryData) and =AVERAGE(SalaryData).

WHERE TO FIND IT

Select all salary values. Formulas tab → Defined Names → Define Name → type SalaryData → OK.

HOW YOU'LL KNOW IT WORKED

Formulas tab → Name Manager shows SalaryData pointing to the salary column range.

E3 — Manage Tables & Table Data

14 points (6 tasks)

2

Task 17 — Convert the employee data range to an Excel Table

WHY THIS MATTERS

Excel Tables auto-expand when you add employees and make filtering and sorting much easier.

WHERE TO FIND IT

Click any cell in the employee data. Insert → Table → confirm range → check My table has headers → OK.

HOW YOU'LL KNOW IT WORKED

The data has alternating row colors and filter dropdown arrows in the header row.

2

Task 18 — Apply a green Medium table style

WHY THIS MATTERS

Green differentiates this HR workbook from financial workbooks that use blue.

WHERE TO FIND IT

Click inside the table. Table Design → Table Styles gallery → find a green Medium style → click it.

HOW YOU'LL KNOW IT WORKED

The table uses a green color scheme with styled header row.

2

Task 19 — Add a Total Row showing average salary and total headcount

WHY THIS MATTERS

Average salary and headcount are the two most common HR dashboard metrics.

WHERE TO FIND IT

Click inside the table. Table Design → Total Row checkbox. Set salary column to Average, name column to Count.

HOW YOU'LL KNOW IT WORKED

A Total row appears showing average salary and employee count.

2

Task 20 — Filter the table to show only Engineering department employees

WHY THIS MATTERS

Department filtering lets HR view a single department's data for reporting.

WHERE TO FIND IT

Click the Department column dropdown → uncheck Select All → check Engineering → OK.

HOW YOU'LL KNOW IT WORKED

Only Engineering employees are visible.

2

Task 21 — Add a new "Bonus Eligible" column to the right of the table

WHY THIS MATTERS

Excel Tables automatically include adjacent columns, extending all table formatting.

WHERE TO FIND IT

Click the first empty cell to the right of the last column header. Type Bonus Eligible and press Enter.

HOW YOU'LL KNOW IT WORKED

The new column adopts the table formatting and is included within the table boundary.

4

Task 22 — Remove duplicate employee records from the table

WHY THIS MATTERS

Duplicate employee records cause double-counting of headcount and payroll totals.

WHERE TO FIND IT

Click inside the table. Table Design → Tools → Remove Duplicates → select columns → OK.

HOW YOU'LL KNOW IT WORKED

Excel confirms 1 duplicate removed. The red duplicate row is gone.

E4 — Formulas & Functions

32 points (12 tasks)

2

Task 23 — Write =SUM(SalaryData) for total payroll cost

WHY THIS MATTERS

Total payroll is the most critical HR budget figure.

WHERE TO FIND IT

Click the Total Payroll cell in the Summary sheet.

FORMULA SYNTAX

```
=SUM(SalaryData)
```

HOW YOU'LL KNOW IT WORKED

The cell shows the total of all 30 salaries.

2

Task 24 — Write =AVERAGE(SalaryData) for average employee salary

WHY THIS MATTERS

Average salary is a benchmark metric for compensation reviews and market comparisons.

WHERE TO FIND IT

Click the Average Salary cell.

FORMULA SYNTAX

```
=AVERAGE(SalaryData)
```

HOW YOU'LL KNOW IT WORKED

The cell shows the average salary across all 30 employees.

2

Task 25 — Write =MAX(SalaryData) for highest salary and =MIN(SalaryData) for lowest

WHY THIS MATTERS

MAX and MIN show the salary range — essential for compensation band analysis.

WHERE TO FIND IT

Click Highest Salary cell → MAX formula. Click Lowest Salary cell → MIN formula.

FORMULA SYNTAX

```
=MAX(SalaryData) and =MIN(SalaryData)
```

HOW YOU'LL KNOW IT WORKED

Highest shows $98,000. Lowest shows $51,000.

2

Task 26 — Write =COUNTA(Employees!B3:B32) to count total number of employees

WHY THIS MATTERS

COUNTA counts non-empty cells — it works on text (names) unlike COUNT which only counts numbers.

WHERE TO FIND IT

Click the Total Employees cell in the Summary sheet.

FORMULA SYNTAX

```
=COUNTA(Employees!B3:B32)
```

HOW YOU'LL KNOW IT WORKED

The cell shows 30.

3

Task 27 — Write IF: if Performance Rating >= 4 show "Bonus Eligible", else "Standard"

WHY THIS MATTERS

IF automates classification decisions for all 30 employees instantly.

WHERE TO FIND IT

Click the first Bonus Eligible cell in the Employees sheet (column J, row 3). Copy down for all employees.

FORMULA SYNTAX

```
=IF(G3>=4,"Bonus Eligible","Standard")
```

HOW YOU'LL KNOW IT WORKED

Employees with rating 4.0+ show "Bonus Eligible". Others show "Standard".

4

Task 28 — Write nested IF for salary band: >=90000="Band A", >=70000="Band B", else "Band C"

WHY THIS MATTERS

Nested IFs test multiple conditions in sequence. Salary banding is standard HR classification.

WHERE TO FIND IT

Click the first Salary Band cell. Copy down for all employees.

FORMULA SYNTAX

```
=IF(F3>=90000,"Band A",IF(F3>=70000,"Band B","Band C"))
```

HOW YOU'LL KNOW IT WORKED

Salaries $90,000+ show Band A. $70,000-$89,999 show Band B. Below $70,000 show Band C.

3

Task 29 — Write COUNTIF to count employees in each department

WHY THIS MATTERS

Department headcount shows where employees are concentrated in the organization.

WHERE TO FIND IT

Click each department headcount cell in the Summary sheet.

FORMULA SYNTAX

```
=COUNTIF(Employees!C3:C32,"Engineering")
```

HOW YOU'LL KNOW IT WORKED

Engineering=8, Sales=6, HR=4, Finance=6, Operations=5 (total=30 — adjust if your column letter differs).

3

Task 30 — Write SUMIF to total salaries by department

WHY THIS MATTERS

Department salary totals show where the payroll budget is being spent.

WHERE TO FIND IT

Click each department salary cell in the Summary sheet.

FORMULA SYNTAX

```
=SUMIF(Employees!C3:C32,"Engineering",Employees!F3:F32)
```

HOW YOU'LL KNOW IT WORKED

Each department shows its total salary spend. All departments combined equal SUM(SalaryData).

3

Task 31 — Use LEFT to extract the first 3 characters of each EmpID

WHY THIS MATTERS

LEFT extracts characters from the start of a text string — used to extract prefixes or codes from structured IDs.

WHERE TO FIND IT

Click the first EmpID Code cell.

FORMULA SYNTAX

```
=LEFT(Employees!A3,3)
```

HOW YOU'LL KNOW IT WORKED

E001 returns "E00". E002 returns "E00".

2

Task 32 — Use TODAY() to calculate years of service from each employee's Start Date

WHY THIS MATTERS

TODAY() minus Start Date divided by 365 gives approximate years of service — an important HR metric.

WHERE TO FIND IT

Click the Years of Service cell.

FORMULA SYNTAX

```
=ROUND((TODAY()-Employees!E3)/365,1)
```

HOW YOU'LL KNOW IT WORKED

The cell shows a decimal like 6.8 representing years. Updates automatically each day.

4

Task 33 — Write XLOOKUP to pull the department manager name from the Departments sheet

WHY THIS MATTERS

XLOOKUP is the modern replacement for VLOOKUP — searches any direction and handles errors more clearly.

WHERE TO FIND IT

Click the first Manager cell in the Employees sheet (column I, row 3). Copy down for all 30 employees.

FORMULA SYNTAX

```
=XLOOKUP(C3,Departments!$A$3:$A$7,Departments!$B$3:$B$7)
```

HOW YOU'LL KNOW IT WORKED

Each employee shows their department manager's name.

4

Task 34 — Wrap the XLOOKUP in IFERROR to show "No Manager" if not found

WHY THIS MATTERS

IFERROR catches any errors the XLOOKUP might return and replaces them with a readable message.

WHERE TO FIND IT

Edit the XLOOKUP formula and wrap it with IFERROR.

FORMULA SYNTAX

```
=IFERROR(XLOOKUP(C3,Departments!$A$3:$A$7,Departments!$B$3:$B$7),"No Manager")
```

HOW YOU'LL KNOW IT WORKED

All employees show a manager name. Missing departments show "No Manager".

E5 — Manage Charts

18 points (6 tasks)

3

Task 35 — Create a Column chart showing employee headcount by department

WHY THIS MATTERS

A column chart comparing department sizes gives HR a visual snapshot of organizational structure.

WHERE TO FIND IT

Select department names and headcount totals from Summary sheet. Insert → Column or Bar Chart → Clustered Column.

HOW YOU'LL KNOW IT WORKED

A column chart appears with one bar per department.

2

Task 36 — Move the headcount chart to its own sheet named "Headcount Dashboard"

WHY THIS MATTERS

A dedicated chart sheet presents the chart at full-page size for executive reports.

WHERE TO FIND IT

Click the chart. Chart Design → Move Chart → New sheet → type "Headcount Dashboard" → OK.

HOW YOU'LL KNOW IT WORKED

A new "Headcount Dashboard" sheet tab appears with the chart at full size.

3

Task 37 — Change the salary distribution chart type from Column to Bar

WHY THIS MATTERS

Bar charts work better for many categories — department names fit better as horizontal labels.

WHERE TO FIND IT

Click the salary chart. Chart Design → Change Chart Type → Bar → Clustered Bar → OK.

HOW YOU'LL KNOW IT WORKED

The chart changes orientation — bars now run horizontally.

3

Task 38 — Add chart title "Average Salary by Department" and axis titles

WHY THIS MATTERS

Axis titles explain what the numbers represent without requiring the reader to guess.

WHERE TO FIND IT

Click chart. Chart Design → Add Chart Element → Chart Title → Above Chart. Then Add Chart Element → Axis Titles → add both axes.

HOW YOU'LL KNOW IT WORKED

The chart has a title at the top and labeled axes.

3

Task 39 — Apply Quick Layout 1 and Chart Style 8 to the department chart

WHY THIS MATTERS

Quick Layouts rearrange chart elements in predefined patterns. Chart Styles apply coordinated color schemes.

WHERE TO FIND IT

Click chart. Chart Design → Quick Layout → Layout 1. Then Chart Design → Chart Styles → Style 8.

HOW YOU'LL KNOW IT WORKED

The chart layout and color scheme update.

4

Task 40 — Add descriptive alt text to all charts in the workbook

WHY THIS MATTERS

Alt text is required for accessible workplace documents.

WHERE TO FIND IT

Right-click each chart → Edit Alt Text → type a description of what the chart shows.

HOW YOU'LL KNOW IT WORKED

Every chart has alt text visible in the Alt Text pane when right-clicked.